
ST-SIMDIFF: BALANCING SPATIOTEMPORAL SIMILARITY AND DIFFERENCE FOR EFFICIENT VIDEO UNDERSTANDING WITH MLLMs

Anonymous authors
Anonymous submission

ABSTRACT

Multimodal Large Language Models (MLLMs) incur substantial computational overhead when videos are represented as long sequences of visual tokens, a challenge that becomes more pronounced as video duration, resolution, and downstream reasoning demands increase. Existing efficiency methods usually reduce this burden by pruning or merging tokens according to importance or similarity. Although these strategies are effective for many redundant regions, they leave a broader aspect of video representation underexplored: changes, transitions, and turning points that may be sparse in token count but central to understanding events across domains. They also often treat spatial and temporal relations separately rather than modeling their joint structure. To address this gap, we frame video token reduction around a general principle: similarity supports redundancy removal, whereas difference supports event preservation. Following this principle, we propose ST-SimDiff, a training-free framework that constructs a spatio-temporal graph over visual tokens to model their local spatial and temporal associations in a unified form. ST-SimDiff then applies a parallel dual-selection strategy: 1) similarity-based selection uses community detection to retain representative tokens for stable or repetitive content; and 2) temporal-difference-based selection locates content-changing points and preserves tokens associated with dynamic shifts. This design keeps both persistent scene information and temporally distinctive evidence under a compact token budget. Experiments on established video understanding benchmarks show that ST-SimDiff outperforms state-of-the-art token compression approaches while substantially reducing computational cost. Our code will be released upon acceptance.

1 INTRODUCTION

The rise of Large Language Models (LLMs) has accelerated the development of Large Vision-Language Models (LVLMs), which now support a wide range of image and video understanding tasks (Zhang et al., 2024c; Liu et al., 2024c). In common video-processing pipelines, an LVLM samples a video into frames and converts each frame into hundreds or thousands of visual tokens before sending them to the language model (Bai et al., 2025; Li et al., 2024). This representation is flexible and broadly applicable, but its cost scales rapidly with video duration and resolution. As a result, token-heavy video inputs create substantial computation and memory burdens, limiting the use of LVLMs in settings that require long-context analysis, responsive interaction, or deployment under constrained hardware budgets (Fu et al., 2024b).

A growing body of work improves LVLM efficiency by reducing the number of visual tokens. These methods can be broadly grouped into two families. Importance-based pruning exploits the observation, highlighted by FastV (Chen et al., 2024), that attention distributions across LVLM layers contain redundancy; tokens with lower attention scores in deeper layers can therefore be removed to reduce computation. Similarity-based merging or selection instead leverages high feature similarity among visual tokens, either across neighboring video frames or within an individual frame. For example, FrameFusion (Fu et al., 2024b) merges similar tokens from adjacent frames, while VisionZip (Yang et al., 2025) selects dominant tokens at the vision-encoder level. Together, these directions

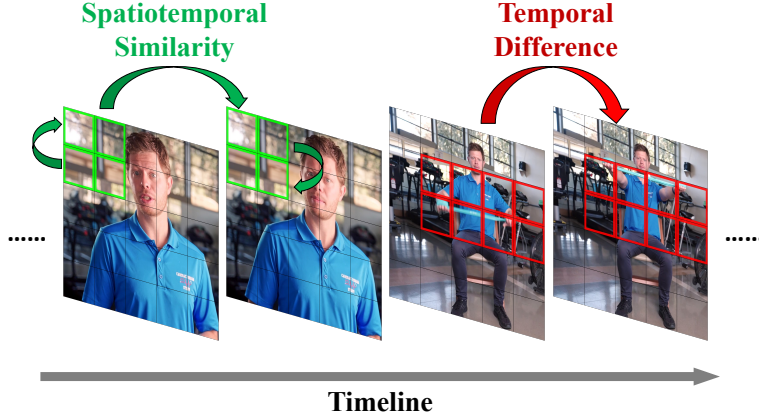


Figure 1: Core motivation of ST-SimDiff. Efficient video understanding should account for two broadly occurring structures: Spatiotemporal Similarity (left), which supports compression of redundant information within a frame or temporal correlations at corresponding positions, but do not jointly analyze spatio-temporal relationships among tokens. This separation can make it difficult to represent dynamic visual patterns whose redundancy and semantics span both space and time. Second, these methods tend to prioritize informational commonality, such as similarity or attention-based importance, while giving less explicit treatment to differences. Since many video tasks depend on recognizing transitions, action onsets, scene changes, or other turning events, a compression method that only seeks repeated evidence may smooth over the very moments needed for correct interpretation.

show that video token budgets can be reduced, but they primarily emphasize common or repeated information.

Despite their progress in visual token reduction, existing approaches have two limitations that can affect diverse video-understanding scenarios. First, many pruning or merging strategies consider spatial correlations within a frame or temporal correlations at corresponding positions, but do not jointly analyze spatio-temporal relationships among tokens. This separation can make it difficult to represent dynamic visual patterns whose redundancy and semantics span both space and time. Second, these methods tend to prioritize informational commonality, such as similarity or attention-based importance, while giving less explicit treatment to differences. Since many video tasks depend on recognizing transitions, action onsets, scene changes, or other turning events, a compression method that only seeks repeated evidence may smooth over the very moments needed for correct interpretation.

We therefore adopt a broader view of efficient video representation: similarity should identify what can be compressed, while difference should identify what must remain visible. Under this view, an effective video token compression algorithm should simultaneously summarize stable content with few tokens and preserve the changes that distinguish one event from another. To this end, we propose **ST-SimDiff**, a dual token selection framework built on a spatio-temporal graph. The framework contains two complementary components. The first, **similarity-based representative token selection**, treats all video tokens as graph nodes and connects them according to feature similarity in the spatio-temporal neighborhood. Graph community detection identifies tightly connected token groups that correspond to stable or recurring visual elements, and central tokens from each group are retained as representatives. The second, **difference-based event token selection**, focuses on temporal graph edges. When the similarity between corresponding tokens in adjacent frames drops sharply, the method treats this change as a transition and retains the associated token as event evidence. The final token set merges representative and event tokens before input to the LLM. Our contributions can be summarized as follows:

- We propose a general token-compression perspective for VLM efficiency research that assigns explicit roles to both similarity and difference.
- We design a spatio-temporal graph framework that jointly models relationships among video tokens and supports a dual selection strategy for representative content and pivotal event evidence.
- We conduct experiments on multiple video understanding benchmarks and base models, showing that ST-SimDiff substantially reduces visual token budgets while maintaining or improving performance under the evaluated conditions.

2 RELATED WORK

2.1 LARGE VISION LANGUAGE MODELS

The rapid advancement of Large Language Models (LLMs) (Touvron et al., 2023; Achiam et al., 2023; Ouyang et al., 2022) has broadly expanded multimodal modeling, giving rise to many Large Vision Language Models (LVLMs) (Liu et al., 2024b; Bai et al., 2023). A common LVLM architecture converts images or sampled video frames into visual tokens through a pre-trained visual encoder and then feeds these tokens to an LLM. Alignment modules such as MLPs (Liu et al., 2024a;b) and Q-Formers (Li et al., 2023; Dai et al., 2023) bridge the vision encoder and language model. This design has enabled strong performance across multimodal tasks (Xi et al., 2025; Holdaway et al., 2024; Zhang et al., 2025a), and in video understanding it has shown promise for longer, higher-resolution, and more complex inputs (Bai et al., 2025; Li et al., 2024; Zhang et al., 2024c; Liu et al., 2024c).

The same token-based interface, however, creates a general efficiency bottleneck for video. Because videos contain continuous frame sequences rather than a single static image, the number of visual tokens can grow far beyond the image setting and place heavy demands on both training and inference. Efficiently processing large video token sequences while retaining task-relevant evidence therefore remains a central problem for LVLM research and for broader deployment scenarios.

2.2 VISUAL TOKEN COMPRESSION

Motivated by the high cost of video inputs, visual token compression has become a major strategy for improving LVLM efficiency. Existing methods mainly compress tokens through importance estimation, similarity analysis, or hybrids of the two.

Earlier work primarily studied importance-based pruning, where tokens judged to contribute less to model predictions are removed. Such methods often use attention scores or feature-based criteria to quantify token importance, as in FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b). Later approaches (Shang et al., 2024; Ye et al., 2025) further incorporate token similarity to reduce redundant evidence while retaining useful information. These methods recognize that even visually important tokens can be repetitive. For instance, FrameFusion (Fu et al., 2024b) combines similarity-based merging with importance-based pruning, VisionZip (Yang et al., 2025) selects dominant tokens while merging contextual information, and VisPruner (Zhang et al., 2025b) uses visual cues to remove redundant items.

Although these studies make substantial progress on visual token redundancy, they generally do not explicitly model the global pattern of spatio-temporal similarity among video tokens. In videos, redundancy can arise within a frame, across adjacent frames, and across moving semantic content. Existing similarity-driven methods such as FrameFusion mainly emphasize temporal similarity between adjacent frames, leaving more complex spatio-temporal structure less fully exploited. This limits compression in strict token-budget settings and motivates a framework that treats similarity and temporal difference as complementary signals for preserving broadly useful video evidence.

3 PROBLEM DEFINITION

Given a video V and a text query Q , a standard Large Vision-Language Model (LVLM) encodes the video as a complete sequence of N visual tokens, $T_{\text{full}} = \{t_1, t_2, \dots, t_N\}$. This token sequence provides a general interface for video-language reasoning, but it is computationally expensive because the self-attention complexity grows quadratically with the token count N , i.e., $O(N^2)$. We therefore study an efficient token selection function, $f(\cdot)$, that receives the full sequence T_{full} and a compression ratio r and outputs a smaller subset $T_{\text{sub}} = f(T_{\text{full}}, r)$. The goal is to preserve downstream LVLM performance with this subset while satisfying the budget constraint that $|T_{\text{sub}}|$ is approximately $r \cdot N$. The central challenge is to define $f(\cdot)$ so that the retained subset T_{sub} captures both representative visual content and event-specific changes under a limited token budget.

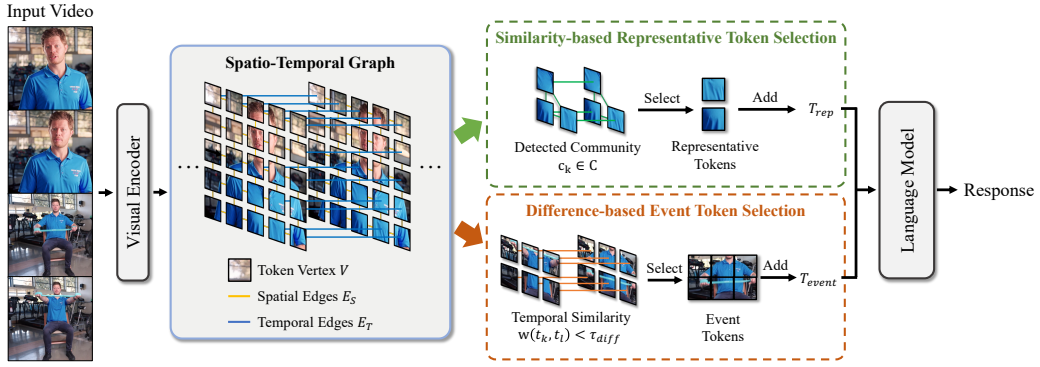


Figure 2: Overview of ST-SimDiff. The framework applies Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection, and Difference-based Event Token Selection to produce a compact token set that preserves both stable content and transition evidence.

4 ST-SIMDIFF

4.1 METHOD OVERVIEW

As illustrated in Figure 2, ST-SimDiff contains two parallel components operating on a spatio-temporal graph constructed from video visual tokens: Similarity-based Representative Token Selection (SRTS) and Difference-based Event Token Selection (DETS). These two components target complementary structures that commonly appear across video inputs, namely persistent similarity and localized temporal change. First, SRTS condenses stable or redundant content. By applying community detection to the graph, it groups highly similar tokens into clusters that can correspond to recurring visual elements, such as static backgrounds or persistent objects. A small number of high-centrality tokens are then selected from each cluster to form the representative set T_{rep} . Second, DETS preserves dynamic shifts. It analyzes temporal graph edges and identifies moments where the similarity between corresponding tokens in adjacent frames drops sharply. Such high-difference points indicate potential transitions, and the tokens around these transitions are retained as event tokens, forming the set T_{trans} . The final subset $T_{sub} = T_{rep} \cup T_{trans}$ combines these two sources of evidence, yielding a compact representation that retains stable scene information while keeping tokens associated with key changes.

4.2 SPATIO-TEMPORAL GRAPH CONSTRUCTION

Given a video, the vision encoder transforms it into a token sequence $T = \{t_1, t_2, \dots, t_N\}$. Each token $t_k \in T$ corresponds to a spatio-temporal location and has a feature vector $\mathbf{x}_k$. We denote the location of t_k by its frame index $T(t_k)$, spatial height index $H(t_k)$, and spatial width index $W(t_k)$.

We represent relationships among tokens with a spatio-temporal graph $G = (V, E)$, where the vertex set V contains all tokens in T . The edge set E is composed of two subsets, $E = E_S \cup E_T$, which encode spatial adjacency and temporal continuity respectively. The spatial edges E_S connect tokens that are adjacent within the same frame:

$$E_S = \{(v_i, v_j) \in V \times V \mid T(t_i) = T(t_j) \text{ and } |H(t_i) - H(t_j)| + |W(t_i) - W(t_j)| = 1\} \quad (1)$$

The temporal edges E_T connect tokens at the same spatial position in adjacent frames, capturing local temporal continuity:

$$E_T = \{(v_i, v_j) \in V \times V \mid H(t_i) = H(t_j), W(t_i) = W(t_j), \text{ and } |T(t_i) - T(t_j)| = 1\} \quad (2)$$

For any edge $w(v_i, v_j) \in E$, its weight is the cosine similarity between the feature vectors of the corresponding tokens:

$$w(v_i, v_j) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|} \quad (3)$$

This sparse graph encodes local spatial relationships and temporal continuity in a single structure, giving both selection modules a shared basis for analyzing redundancy and change.

4.3 SIMILARITY-BASED REPRESENTATIVE TOKEN SELECTION

Video content often contains substantial spatio-temporal redundancy. A static scene may produce background tokens that remain similar over time, while a moving object can preserve semantic similarity even as its spatial location changes. Methods that treat spatial and temporal similarity separately can miss this broader structure. Our spatio-temporal graph is designed to capture joint similarities so that semantically related tokens can form dense “communities” or “clusters” based on feature similarity, regardless of their exact frame or location. Selecting representatives from these communities provides a general way to compress repeated visual evidence while preserving core semantics.

To identify strongly correlated token clusters, we first threshold the graph G . We set a similarity threshold τ_{sim} and retain only edges whose weights exceed this threshold, forming a new graph $G' = (V, E')$, where $E' = \{(v_i, v_j) \in E \mid w(v_i, v_j) > \tau_{\text{sim}}\}$. Next, we apply a graph community detection algorithm (e.g., the Louvain method (Blondel et al., 2008)) on G' to identify token clusters $C = \{c_1, c_2, \dots, c_m\}$. For each community $c_k \in C$, we rank and filter its tokens according to centrality. The centrality score $S_c(t_a)$ of token $t_a \in c_k$ is defined as its average similarity to all other tokens in the same community:

$$S_c(t_a) = \frac{1}{|c_k| - 1} \sum_{t_b \in c_k, b \neq a} w(t_a, t_b) \quad (4)$$

We then set the intra-community retention rate to the global compression ratio r and keep the top $\lceil |c_k| \cdot r \rceil$ tokens with the highest centrality scores from each community c_k . This filtering rule applies uniformly across communities, including single-node communities that correspond to unique tokens, for which the sole node is retained. Formally, the representative token set T_{rep} is defined as:

$$T_{\text{rep}} = \bigcup_{k=1}^m \text{TopK}_{t \in c_k, \text{score}=S_c(t)} (\lceil |c_k| \cdot r \rceil) \quad (5)$$

where the TopK function returns the set of k elements with the highest scores from a given set. This strategy compresses stable and persistent video content by retaining the most central evidence within each semantic cluster.

4.4 DIFFERENCE-BASED EVENT TOKEN SELECTION

If similarity captures the regular structure of a video, difference captures deviations that may define events. A compression method that focuses only on similarity can preserve what is present but may underrepresent what happened. Across many video tasks, the answer-relevant evidence may be a new object entering the scene, an action beginning or ending, or a scene transition. In token space, these changes appear as sharp differences between temporally adjacent features. Capturing such turning points is therefore important for questions about timing, causality, and event progression.

Temporal difference is especially useful because major visual changes often alter corresponding positions in adjacent frames. We therefore analyze the temporal edges (E_T) in the spatio-temporal graph. For any temporal edge $(v_k, v_l) \in E_T$ connecting temporally adjacent tokens t_k and t_l , we set a dynamic threshold τ (e.g., the 95th percentile of all temporal edge difference scores). When the difference score of (v_k, v_l) exceeds this threshold, i.e., $w(t_k, t_l) < \tau_{\text{diff}}$, we treat the later token t_l as a critical event token and retain it. Formally, the event token set T_{event} is defined as:

$$T_{\text{event}} = \{t_l \mid \exists t_k \text{ s.t. } (v_k, v_l) \in E_T, \\ T(t_l) > T(t_k), \text{ and } w(v_k, v_l) < \tau_{\text{diff}}\} \quad (6)$$

This selection preserves tokens that mark significant temporal changes. By retaining transition evidence alongside representative content, the compression process reduces the chance that dynamic aspects of the video narrative are removed.

4.5 OVERALL REDUCTION PROCESS

In ST-SimDiff, we compute the representative token set T_{rep} and the event token set T_{event} in parallel and take their union, $T_{\text{candidate}} = T_{\text{rep}} \cup T_{\text{event}}$, as the initial candidate set. To meet the target token count $N_{\text{target}} = \lceil r \cdot N \rceil$, we add a final pruning step. If the candidate set size $|T_{\text{candidate}}|$ exceeds N_{target} , we remove the $|T_{\text{candidate}}| - N_{\text{target}}$ tokens with the lowest importance. Following existing work (Chen et al., 2024), token importance is determined by the attention score in the shallow layers of the LLM. This step lets the method satisfy different computational budgets while retaining the graph-selected evidence most relevant under that budget.

Overall, ST-SimDiff combines graph-based token selection with attention-based dynamic pruning. The graph component uses the intrinsic structure of video content, including both similarity and difference, to preserve representative and transitional evidence. The pruning component provides a budget-control mechanism that can adapt the selected set to a prescribed compression ratio. This combination yields a principled and flexible compression process that balances efficiency with downstream video understanding performance.

4.6 COMPUTATIONAL COMPLEXITY ANALYSIS

The computational overhead of ST-SimDiff can be analyzed through three stages: Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection (SRTS), and Difference-based Event Token Selection (DETS).

Graph construction requires computing cosine similarity between each of the N visual tokens and its spatially and temporally adjacent neighbors. Because each token has only a small constant number of neighbors, this stage is a single pass over the token sequence with complexity $O(Nd)$, where d is the feature dimension.

After graph construction, SRTS identifies token clusters. As described in our implementation details (Section 5.2), we use a connected components algorithm, which has near-linear time complexity $O(N + |E|)$. Since the number of edges $|E|$ is at most $3N$, this reduces to $O(N)$. The intra-community filtering step could be expensive if all-pairs similarity were computed inside each community c_k , with naive complexity $O(|C| \cdot |c_k|^2 \cdot d)$. To avoid this cost, we constrain the maximum community size. If a community exceeds a predefined threshold, such as $|c_k| > \sqrt{N}$, we partition it so that filtering remains efficient and does not dominate the total cost. Although graph clustering algorithms such as Louvain and Leiden support richer community definitions, their complexity is higher and can approach $O(N \log N)$. We therefore use the simpler connected components algorithm in practice to keep the compression overhead low.

The DETS module performs a single pass over all temporal edges and identifies significant changes by comparing token similarities with the threshold. This stage has linear complexity $O(Nd)$.

Taken together, the dominant operations in ST-SimDiff are linear in the number of tokens, giving an overall complexity of $O(Nd)$. This is substantially cheaper than the $O(N^2d)$ self-attention cost of the Large Language Model. Thus, under the studied setting, ST-SimDiff provides token reduction with a small additional overhead relative to the main inference computation.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Baselines To evaluate ST-SimDiff beyond a single backbone, we use LLaVA-Video (Zhang et al., 2024c) and NVILA (Liu et al., 2024c) as base models and compare against several state-of-the-art video token compression techniques. The baselines span multiple compression principles, including **importance-based reduction methods** such as FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b), as well as **hybrid reduction methods** such as VisionZip (Yang et al., 2025), FrameFusion (Fu et al., 2024b), and PruMerge (Shang et al., 2024). For calibration, we also include the original uncompressed model (Vanilla) as a full-token upper bound and Random Sampling with the same token count as a lower-bound reference.

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
LLaVA-Video	-	-	-	63.3	-	-	58.2	57.3
Token Retain Ratio $r = 30\%$								
FastV	68.2	58.6	51.2	59.3	48.0	59.9	53.5	51.3
PruMerge	70.2	57.3	52.2	59.9	49.6	60.5	54.7	50.9
FasterVLM	71.3	57.6	51.4	60.1	51.0	61.3	55.8	52.6
VisionZip	68.4	55.9	50.4	58.3	46.9	60.3	53.2	53.0
FrameFusion	74.0	59.8	50.0	61.3	49.7	63.3	56.0	53.0
ST-SimDiff (Ours)	74.7	61.9	53.0	63.2	52.5	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$								
FastV	73.9	61.4	51.3	62.2	50.1	62.2	55.7	54.7
PruMerge	72.3	58.7	52.8	61.3	51.4	63.2	56.9	54.6
FasterVLM	74.0	59.2	51.8	61.7	51.3	62.2	56.4	56.2
VisionZip	72.6	57.7	51.6	60.6	51.3	63.2	56.8	54.2
FrameFusion	74.6	61.2	52.0	62.6	51.7	64.2	57.6	55.8
ST-SimDiff (Ours)	76.2	62.4	52.7	63.8	52.3	64.3	57.9	57.3

Table 1: Performance comparison across long-form video understanding benchmarks using LLaVA-Video-7B with 64 input frames (%). The best result among the evaluated compression methods is emphasized in **bold**.

Benchmarks To test performance across different long-video understanding demands, we adopt three benchmark datasets: VideoMME (Fu et al., 2024a), LongVideoBench (Wu et al., 2025), and EgoSchema (Mangalam et al., 2024). VideoMME covers foundational video understanding with diverse video types and three duration categories: Short, Medium, and Long. LongVideoBench measures retrieval and reasoning over long-video details through a “referring reasoning” task, with videos of four lengths: 15s, 60s, 600s, and 3600s. EgoSchema is a question-answering dataset built around long-term first-person videos, emphasizing character intentions and causal chains of actions.

5.2 IMPLEMENTATION DETAILS

All experiments are run on NVIDIA L20 GPUs with 48GB VRAM under Ubuntu 22.04. Inference evaluation uses the *lmms-eval* (Zhang et al., 2024a) library. For LLaVA-Video, we follow its original 64-frame input setting; to compare computation and resource usage under aligned conditions, we also set NVILA to 64 input frames. Unless otherwise specified, we set the similarity threshold to $\tau_{\text{sim}} = 0.8$ and the difference threshold to $\tau_{\text{diff}} = 0.2$. We evaluate token retention ratios of $r = 30\%$ and $r = 50\%$. In the implementation, we use connected component finding as the community detection algorithm to balance runtime overhead and performance.

5.3 COMPARISON WITH STATE-OF-THE-ARTS

We compare ST-SimDiff with mainstream efficient video language model (VLM) compression methods on VideoMME, LongVideoBench, and EgoSchema. To examine whether the observed behavior transfers across model classes, we report results on two base models, LLaVA-Video-7B and NVILA-8B, under token retention ratios of 30% and 50%. The results are shown in Table 1 and Table 2.

(1) ST-SimDiff consistently outperforms the competing compression methods across the evaluated configurations. At a 50% token retention ratio, its overall performance on both base models sur-

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
NVILA-Video	-	-	-	61.5	-	-	56.3	52.9
Token Retain Ratio $r = 30\%$								
FastV	69.4	55.3	48.8	57.9	50.8	55.6	53.0	49.7
PruMerge	71.0	53.7	49.9	58.2	49.3	58.1	53.4	47.5
FasterVLM	72.6	56.9	51.0	60.1	48.7	57.8	53.0	49.3
VisionZip	71.4	56.2	49.7	59.1	46.3	56.2	50.9	48.9
FrameFusion	72.1	55.7	48.7	58.8	51.3	59.3	54.9	51.3
ST-SimDiff (Ours)	73.3	57.7	49.7	60.2	51.7	59.2	55.2	51.7
Token Retain Ratio $r = 50\%$								
FastV	71.9	58.3	49.3	58.9	49.5	58.8	53.9	50.2
PruMerge	71.2	54.8	46.8	57.6	49.7	58.7	53.9	48.9
FasterVLM	74.6	57.8	50.1	60.8	48.5	58.1	53.0	50.5
VisionZip	72.8	58.1	50.6	60.5	50.4	58.9	54.4	50.3
FrameFusion	72.7	57.2	48.3	59.4	51.0	59.2	54.8	52.6
ST-SimDiff (Ours)	73.9	59.7	51.4	61.7	52.3	61.3	56.5	52.5

Table 2: Performance comparison across long-form video understanding benchmarks using NVILA-Video-8B with 64 input frames (%). The best result among the evaluated compression methods is emphasized in **bold**.

passes the other compression algorithms and, on some benchmarks, matches or exceeds the corresponding full-token model. This pattern suggests that the similarity-difference selection principle can remove redundant tokens without discarding task-relevant evidence under the tested conditions.

(2) Among importance-based baselines such as FastV and FasterVLM, FasterVLM generally performs more strongly. Their shared limitation is that important tokens can still be temporally repetitive, so retaining them may leave video redundancy insufficiently compressed. Hybrid methods such as PruMerge, VisionZip, and FrameFusion are more competitive than importance-only methods, with FrameFusion usually providing the strongest baseline performance. However, these methods still largely optimize for representative “commonality” information, which can leave transition-driven evidence, such as “turning points” and “abrupt events”, less explicitly protected.

(3) Overall, the baselines show that video compression increasingly recognizes redundancy, but two broader challenges remain: jointly modeling spatio-temporal similarity among visual tokens, and explicitly detecting differences that mark key changes. ST-SimDiff addresses both through a unified spatio-temporal graph and a difference-aware selection path, which together explain the gains observed across the evaluated benchmarks and backbones.

5.4 ABLATION STUDY

To isolate how each component contributes to the broader compression behavior of ST-SimDiff, we conduct ablation studies over its two selection paths. Starting from a Baseline strategy that uses only basic importance-based pruning, we progressively add the similarity selection module (+Sim) and the difference selection module (++Diff). All ablations use token retention ratios of 30% and 50%, with results reported in Table 5.

(1) The +Sim module consistently improves over the baseline, indicating that similarity modeling is useful for compressing repeated visual evidence. We further decompose similarity modeling into spatial-only (Spatial), temporal-only (Temporal), and joint spatio-temporal (Spa. + Tem.) variants.

Method	VideoMME	LongVideoBench	EgoSchema
Token Retain Ratio $r = 30\%$			
Baseline	60.3	56.2	54.8
+ Sim	Spatial	61.5	56.5
	Temporal	61.7	56.8
	Spa. + Tem.	62.6	57.0
++ Diff (Proposed)	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$			
Baseline	63.2	56.7	56.5
+ Sim	Spatial	63.3	57.0
	Temporal	63.7	57.2
	Spa. + Tem.	63.7	57.8
++ Diff (Proposed)	63.8	57.9	57.3

Table 3: Component ablation results for similarity- and difference-based selection modules (%).

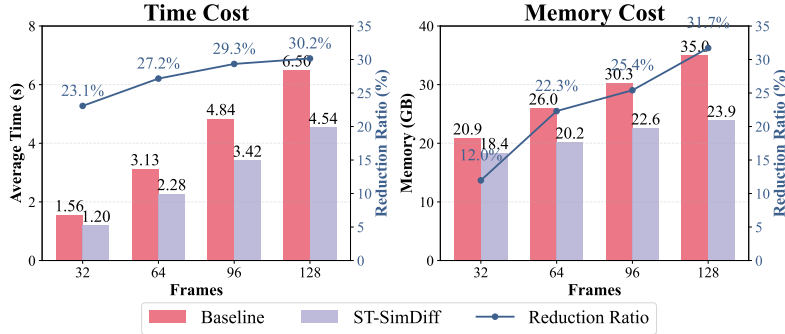


Figure 3: Computational cost comparison between ST-SimDiff and the baseline LLaVA-Video model, measured by average inference time and peak GPU memory usage under a 30% token budget.

The joint variant performs best, supporting the view that redundancy in video is not purely spatial or temporal but often spans both dimensions.

(2) After adding the joint spatio-temporal similarity module, we introduce the difference selection module (++Diff) to obtain the complete framework. The additional gain is especially clear at the 30% retention ratio, where compression is more aggressive and event tokens are easier to lose. In this setting, ++Diff complements +Sim by preserving distinctive transition tokens rather than only representative redundant content. At the 50% ratio, the marginal gain is smaller because the larger retained set is more likely to already include these tokens and performance is closer to the full-token upper bound. This ablation supports the paper’s central framing: robust video token compression benefits from balancing similarity-based redundancy reduction with difference-based event preservation.

5.5 COMPUTATIONAL COST ANALYSIS

To quantify the practical efficiency of ST-SimDiff, we measure inference time and peak GPU memory usage under a 30% token budget. We compare against the uncompressed baseline model (Baseline LLaVA-Video). Figure 3 reports results for input lengths ranging from 32 to 128 frames, covering different video-token loads within the same evaluation setup.

Inference Time ST-SimDiff reduces inference time across the tested frame lengths. As input length increases, the baseline model’s average inference time rises from 1.56 seconds at 32 frames to 6.50 seconds at 128 frames. ST-SimDiff slows this growth, requiring 4.54 seconds for 128 frames. The time-saving rate therefore increases for longer inputs, from 23.0% at 32 frames to 30.2% at 128 frames.

Memory Cost ST-SimDiff also lowers peak GPU memory usage. The baseline memory footprint increases with video length, from 20.9 GB at 32 frames to 35.0 GB at 128 frames. By reducing the number of visual tokens passed to the large language model, ST-SimDiff lowers the 128-frame peak memory usage to 23.9 GB, saving 31.7% relative to the baseline. This reduction is relevant for longer-video settings and for deployments with limited memory capacity.

In summary, the evaluated results show that ST-SimDiff improves both performance and computational efficiency, reducing time and memory costs while preserving the evidence needed for video understanding.

6 CONCLUSION

This paper addresses the computational inefficiency of Multimodal Large Language Models for video inputs, where large visual-token sequences and unprotected content changes can both limit efficient understanding. We propose ST-SimDiff, a training-free framework that builds a spatio-temporal graph to model token associations and applies a dual selection strategy: similarity-based selection identifies redundant representative content through community detection, while temporal-difference-based selection preserves tokens at key transition points. Experiments on VideoMME, LongVideoBench, and EgoSchema with multiple base models show that ST-SimDiff reduces processing time and memory usage while achieving strong performance among the evaluated compression methods. These results support a broader direction for video representation: efficient token selection should jointly consider content representativeness and transitional events.

REFERENCES

- Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. Qwen-vl: A versatile vision-language model for understanding, localization, text reading, and beyond. *arXiv preprint arXiv:2308.12966*, 1(8), 2023.
- Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, et al. Qwen2. 5-vl technical report. *arXiv preprint arXiv:2502.13923*, 2025.
- Vincent D Blondel, Jean-Loup Guillaume, Renaud Lambiotte, and Etienne Lefebvre. Fast unfolding of communities in large networks. *Journal of statistical mechanics: theory and experiment*, 2008 (10):P10008, 2008.
- Liang Chen, Haozhe Zhao, Tianyu Liu, Shuai Bai, Junyang Lin, Chang Zhou, and Baobao Chang. An image is worth 1/2 tokens after layer 2: Plug-and-play inference acceleration for large vision-language models. In *European Conference on Computer Vision*, pp. 19–35. Springer, 2024.
- Wenliang Dai, Junnan Li, Dongxu Li, Anthony Meng Huat Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale Fung, and Steven Hoi. Instructblip: Towards general-purpose vision-language models with instruction tuning, 2023.
- Chaoyou Fu, Yuhang Dai, Yongdong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, et al. Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal llms in video analysis. *arXiv preprint arXiv:2405.21075*, 2024a.

- Tianyu Fu, Tengxuan Liu, Qinghao Han, Guohao Dai, Shengen Yan, Huazhong Yang, Xuefei Ning, and Yu Wang. Framefusion: Combining similarity and importance for video token reduction on large visual language models. *arXiv preprint arXiv:2501.01986*, 2024b.
- Holdaway et al. Meta-designing quantum experiments with language models. *Nature Machine Intelligence*, 2024. doi: 10.1038/s42256-025-01153-0.
- Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Peiyuan Zhang, Yanwei Li, Ziwei Liu, et al. Llava-onevision: Easy visual task transfer. *arXiv preprint arXiv:2408.03326*, 2024.
- Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International conference on machine learning*, pp. 19730–19742. PMLR, 2023.
- Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 26296–26306, 2024a.
- Haotian Liu, Chunyuan Li, Yuheng Li, Bo Li, Yuanhan Zhang, Sheng Shen, and Yong Jae Lee. Llava-next: Improved reasoning, ocr, and world knowledge, January 2024b. URL <https://llava-vl.github.io/blog/2024-01-30-llava-next/>.
- Zhijian Liu, Ligeng Zhu, Baifeng Shi, Zhuoyang Zhang, Yuming Lou, Shang Yang, Haocheng Xi, Shiyi Cao, Yuxian Gu, Dacheng Li, et al. Nvila: Efficient frontier visual language models. *arXiv preprint arXiv:2412.04468*, 2024c.
- Karttikeya Mangalam, Raiymbek Akshulakov, and Jitendra Malik. Egoschema: A diagnostic benchmark for very long-form video language understanding. *Advances in Neural Information Processing Systems*, 36, 2024.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, 35: 27730–27744, 2022.
- Yuzhang Shang, Mu Cai, Bingxin Xu, Yong Jae Lee, and Yan Yan. Llava-prumerge: Adaptive token reduction for efficient large multimodal models. *arXiv preprint arXiv:2403.15388*, 2024.
- Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023.
- Haoning Wu, Dongxu Li, Bei Chen, and Junnan Li. Longvideobench: A benchmark for long-context interleaved video-language understanding. *Advances in Neural Information Processing Systems*, 37:28828–28857, 2025.
- Haoning Xi, Zhiqi Shao, David A Hensher, John D Nelson, Huaming Chen, and Kasun Wijayaratna. A multi-task transformer with mixture-of-experts for personalized periodic predictions of individual travel behavior in multimodal public transport. *Transportation Research Part C: Emerging Technologies*, 179:105287, 2025.
- Senqiao Yang, Yukang Chen, Zhuotao Tian, Chengyao Wang, Jingyao Li, Bei Yu, and Jiaya Jia. Visionzip: Longer is better but not necessary in vision language models. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 19792–19802, 2025.
- Weihao Ye, Qiong Wu, Wenhao Lin, and Yiyi Zhou. Fit and prune: Fast and training-free visual token pruning for multi-modal large language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pp. 22128–22136, 2025.
- Zhang et al. Solving the many-electron schrödinger equation with a transformer-based framework. *Nature Communications*, 2025a. doi: 10.1038/s41467-025-63219-2.

594 Kaichen Zhang, Bo Li, Peiyuan Zhang, Fanyi Pu, Joshua Adrian Cahyono, Kairui Hu, Shuai Liu,
595 Yuanhan Zhang, Jingkang Yang, Chunyuan Li, and Ziwei Liu. Lmms-eval: Reality check on
596 the evaluation of large multimodal models, 2024a. URL [https://arxiv.org/abs/2407.](https://arxiv.org/abs/2407.12772)
597 12772.

598 Qizhe Zhang, Aosong Cheng, Ming Lu, Zhiyong Zhuo, Minqi Wang, Jiajun Cao, Shaobo Guo,
599 Qi She, and Shanghang Zhang. [cls] attention is all you need for training-free visual token prun-
600 ing: Make vlm inference faster. *arXiv e-prints*, pp. arXiv-2412, 2024b.

601 Qizhe Zhang, Aosong Cheng, Ming Lu, Renrui Zhang, Zhiyong Zhuo, Jiajun Cao, Shaobo Guo,
602 Qi She, and Shanghang Zhang. Beyond text-visual attention: Exploiting visual cues for effective
603 token pruning in vlms. *arXiv preprint arXiv:2412.01818*, 2025b.

604 Yuanhan Zhang, Jinming Wu, Wei Li, Bo Li, Zejun Ma, Ziwei Liu, and Chunyuan Li. Video
605 instruction tuning with synthetic data. *arXiv preprint arXiv:2410.02713*, 2024c.

606
607
608
609
610
611
612
613
614
615
616
617
618
619
620
621
622
623
624
625
626
627
628
629
630
631
632
633
634
635
636
637
638
639
640
641
642
643
644
645
646
647

A VISUALIZATION OF THE TOKEN SELECTION PROCESS

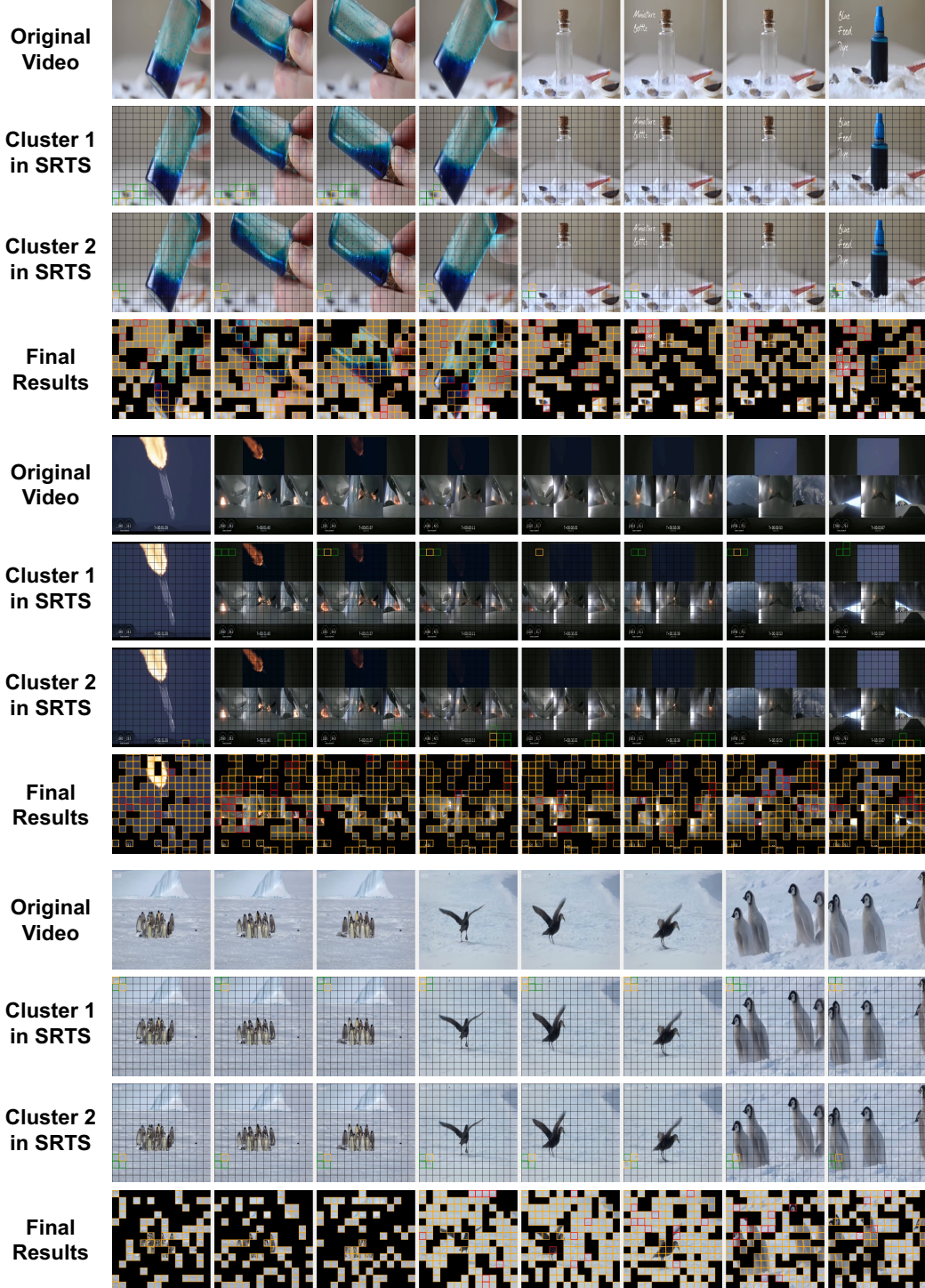


Figure 4: Visualization of the ST-SimDiff process. The visualization decomposes token selection into communities detected by SRTS (shown as grids in Cluster 1 and 2) and event-oriented tokens detected by DETS. The final result illustrates how sparse representative tokens (yellow) preserve stable content while dense event tokens (red) retain dynamic actions.

To make the ST-SimDiff framework more concrete, Figure 4 visualizes intermediate and final token selections on a video sequence involving dynamic object manipulation. For Similarity-based Representative Token Selection (SRTS), the rows labeled “Cluster 1” and “Cluster 2” show how graph community detection groups redundant spatio-temporal regions, including static background areas and stationary objects, into coherent communities. The resulting sparse yellow boxes indicate central “representative tokens” selected from these communities. This example illustrates how SRTS can compress stable visual evidence without requiring all background tokens to be retained.

Difference-based Event Token Selection (DETS) complements this compression by emphasizing dynamic moments. In the “Final Results” row, tokens activated by high temporal difference are marked with dense red bounding boxes. These boxes align with the moving hand and the changing blue liquid, showing that the method identifies “turning points” where temporal difference exceeds the threshold (τ_{diff}). The visualization therefore demonstrates how fine-grained action details and rapid visual changes can be preserved while static evidence is compressed.

The final overlay shows the complementary roles of the two modules. Sparse yellow tokens summarize stable background content, while dense red tokens focus on rapid motion. This contrast visualizes the broader motivation of ST-SimDiff: similarity-based selection removes repeated spatio-temporal evidence, and difference-based selection protects transition evidence that similarity-only strategies may overlook. The example thus illustrates how the framework balances computational efficiency with preservation of critical visual information.

B COMPARISON RESULTS ON QWEN2.5-VL

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
Qwen2.5-VL-64	74.2	62.7	51.8	62.9	54.2	64.8	59.2	63.0
Qwen2.5-VL-128	77.4	65.8	56.1	66.4	56.2	65.1	60.4	63.9
Token Retain Ratio $r = 30\%$								
ST-SimDiff-64	73.8	61.7	51.8	62.4	52.2	62.1	56.8	62.7
ST-SimDiff-128	75.4	64.7	55.2	65.1	53.1	65.0	58.6	64.4
Token Retain Ratio $r = 50\%$								
ST-SimDiff-64	74.2	62.1	52.2	62.9	54.2	63.2	58.4	62.7
ST-SimDiff-128	76.8	66.3	55.8	66.3	54.9	65.3	59.8	64.5

Table 4: Performance comparison across long-form video understanding benchmarks using Qwen2.5-VL-7B with 64 and 128 input frames (%).

To further examine whether the proposed selection principle transfers beyond the main backbones, we conduct additional experiments on Qwen2.5-VL. These evaluations use VideoMME and LongVideoBench and test both 64-frame and 128-frame inputs, allowing us to assess whether the method remains effective across another model family and different input lengths.

The results in Table 4 show that ST-SimDiff remains effective on Qwen2.5-VL. With a 30% token retention ratio, the method achieves 62.4 on VideoMME and 56.8 on LongVideoBench. Increasing the token budget to 50% raises performance to 62.9 on VideoMME and 58.4 on LongVideoBench. At the 50% ratio, VideoMME performance matches the full-token upper bound, while LongVideoBench remains close to its full-performance counterpart. These results support the broader applicability of the framework within the evaluated model and benchmark conditions.

C ABLATION STUDY RESULTS ON NVILA

To examine whether each component behaves similarly across base models, we also conduct ablation studies on NVILA-Video-8B (Liu et al., 2024c). The setup follows the main ablation study: we begin with a Baseline strategy using fundamental importance-based pruning and then progressively add the similarity selection module (+ Sim) and the difference selection module (++ Diff).

Method	VideoMME	LongVideoBench	EgoSchema
Token Retain Ratio $r = 30\%$			
Baseline	59.3	53.8	50.2
+ Sim	Spatial	59.5	54.5
	Temporal	59.6	54.3
	Spa. + Tem.	59.9	54.6
++ Diff (Proposed)	60.2	55.2	51.7
Token Retain Ratio $r = 50\%$			
Baseline	60.9	55.3	51.5
+ Sim	Spatial	61.1	55.7
	Temporal	61.3	55.6
	Spa. + Tem.	61.4	56.0
++ Diff (Proposed)	61.7	56.5	52.5

Table 5: Component ablation results on the NVILA-Video-8B model, evaluating similarity- and difference-based selection modules (%).

As shown in Table 5, the NVILA-Video-8B results follow trends that are consistent with those observed on LLaVA-Video-7B. Adding the similarity module (+ Sim) improves the baseline at both 30% and 50% token retention ratios. Within this module, we compare spatial-only (Spatial), temporal-only (Temporal), and joint spatio-temporal (Spa. + Tem.) strategies. The joint strategy performs best; for example, at $r = 50\%$, it reaches 61.4% on VideoMME and outperforms the spatial-only and temporal-only variants. This supports the claim that video redundancy is more effectively captured when spatial and temporal associations are considered together. After adding the best joint similarity module, we introduce the dissimilarity selection module (++ Diff) to form the complete ST-SimDiff framework. This addition provides the largest gains in the table, showing that the difference path contributes beyond redundancy compression by preserving event-related changes. The component-level pattern on NVILA therefore reinforces the broader idea that efficient video token selection should handle both repeated content and temporal change.

D ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON LLaVA-VIDEO

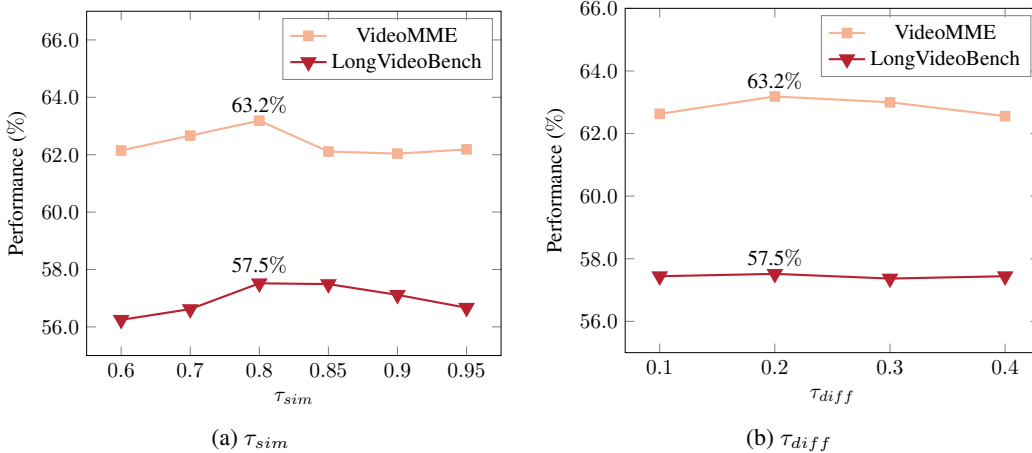


Figure 5: Ablation study results for different values of τ_{diff} and τ_{sim} on VideoMME and LongVideoBench, measuring sensitivity to similarity and difference thresholds.

In ST-SimDiff, the similarity threshold τ_{sim} and the difference threshold τ_{diff} control representative-token and transition-token selection, respectively. We study their effect through the ablations in Figure 6. Performance follows a similar pattern for both parameters, improving up to a stable region and then decreasing, while remaining reasonably robust over nearby values. For τ_{sim} , too low a threshold can make community detection imprecise, whereas too high a threshold can fragment semantic clusters. For τ_{diff} , too low a threshold can introduce noisy transitions, while too high a threshold can miss key events. The best observed setting is $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$, which we use in the other experiments.

E ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON NVILA-VIDEO

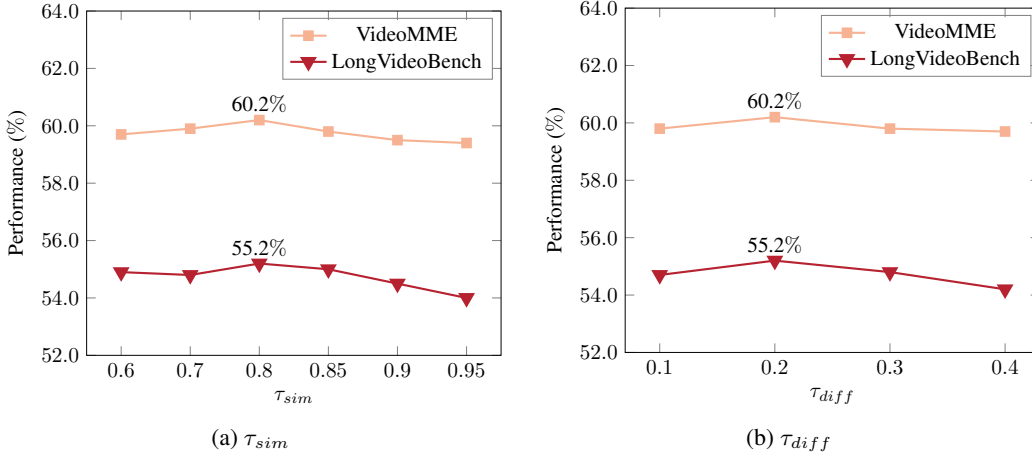


Figure 6: Ablation study results for different values of τ_{diff} and τ_{sim} on NVILA-Video-8B, evaluating threshold sensitivity across two benchmarks.

To assess hyperparameter robustness on another base model, we conduct threshold ablations on NVILA-Video-8B. As shown in Fig. 6, we evaluate how the similarity threshold, τ_{sim} , and the difference threshold, τ_{diff} , affect performance on VideoMME and LongVideoBench.

Impact of Similarity Threshold τ_{sim} . We test τ_{sim} values in the range $[0.6, 0.95]$. The results indicate that performance is relatively insensitive to moderate threshold changes. On VideoMME, the peak value of 60.2% occurs at $\tau_{sim} = 0.8$; on LongVideoBench, the same threshold also yields the best result, 55.2%. Performance decreases slightly when the threshold is too high or too low, but the overall fluctuation is limited, suggesting stable behavior across a broad similarity-threshold range.

Impact of Difference Threshold τ_{diff} . We test τ_{diff} in the range $[0.1, 0.4]$. Similar to τ_{sim} , performance remains robust to nearby values. Both benchmarks reach their best results around $\tau_{diff} = 0.2$, with VideoMME at 60.2% and LongVideoBench at 55.2%. This suggests that a difference threshold near 0.2 effectively captures event-related changes while maintaining token compression.

In summary, these ablations show that ST-SimDiff is not overly sensitive to its key thresholds within the evaluated ranges. The results support using $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$ in the main experiments, as these values consistently perform well across the tested benchmarks and base models.